

When Relevance Does Not Transfer

Consumer-Dependent Retrieval Utility in Sequential Decision Systems

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Abstract. Retrieval is evaluated by relevance, and increasingly by the quality of the text a language model then generates. A growing class of deployments instead feeds retrieved evidence to a decision policy, where an error’s cost depends on which action it induces and on how the system evolves afterwards. The observation that topical relevance can diverge from task value is textbook information retrieval [3], and recent retrieval-augmented generation (RAG) evaluation has measured versions of it: relevance labels correlate weakly with downstream performance [30], grounding utility is model-specific [18], and distracting passages can hurt more than random ones [7,33] — with irrelevant context distracting even strong models [31] while retrieval helps mostly tail queries [24]. Selective retrieval [20,2] and position sensitivity [23] are established remedies our design does not implement. What those studies do not provide is a frozen, regenerable instrument that scores evidence by *sequential realised return* under a fixed controller, with the loss attributed across retrieval, interpretation and control. We build one: an operational report feed, BM25 / dense / hybrid / reranked retrieval, naive and provenance-aware LLM consumers plus a model-free rule-based reference, a development-tuned no-text comparator (labeled extension, outside the frozen matrix), a fixed receding-horizon controller, and a simulator that returns realised operating profit over 160 held-out queries $\times$ 5 paired seeds. The measured pattern is a mismatch, not a scarcity: for LLM consumers, real retrieval reduces return relative to no-text operating points (all C1 arms -41 to -104 and three of eight C3 cells, Holm-robust; remaining real-retriever comparisons directionally negative but Holm-n.s., except random/C3/14B $+11.0$, an abstention-fallback cell), while oracle-selected evidence from the same corpus adds up to $+178$; a relevance-to-factual refinement gap of 0.00 and a downstream control gap of $+456.8$ locate the loss. Relevance-only evaluation cannot see the consumer dependence we measure: a retriever $\times$ consumer interaction that survives Holm correction for rule-vs-LLM contrasts. Every table regenerates from stored episodes; all analysis code, model pins and randomisation are frozen.

Keywords: Retrieval evaluation · Downstream utility · Evidence selection · Consumer dependence · Negative results.

1 Introduction

An operator monitoring a supply network receives a stream of reports. Most concern other sites; some are stale; a few describe conditions that should change what they order this period. Choosing which reports to act on is an information access problem, and its consequences are sequential: an order placed now arrives several periods later, into a state that earlier decisions produced.

Standard retrieval evaluation scores such a system by topical relevance. That topical relevance need not equal task value has been known for decades, from situational relevance to task-based evaluation [3], and the RAG literature has recently quantified single-step versions of the divergence: eRAG shows upstream relevance is a weak proxy for end-task utility [30]; GroGU shows grounding utility differs per consuming model [18]; strong retrievers can retrieve harder distractors than weak ones [7,33]. None of these gives IR a reusable way to score evidence by what it does to a *sequential* decision process with persisting state and asymmetric costs. That instrument is this paper’s contribution; the empirical pattern it reveals is consistent with, and narrower than, the effects above.

We ask when retrieved text improves sequential decisions, and make the whole path executable so the question can be answered rather than argued:

$$q, \mathcal{D} \longrightarrow R_k(q, \mathcal{D}) \longrightarrow b(z \mid \cdot) \longrightarrow \pi(a_t \mid s_t, b) \longrightarrow s_{t+1} \longrightarrow U. \quad (1)$$

A retrieval system R_k selects k documents from a candidate pool $\mathcal{D}$ under a fixed query q ; a fixed consumer maps the selected text to a belief b over operational regimes; a fixed controller maps state and belief to an action; a simulator returns realised utility U . Holding everything after retrieval constant — with shared seeds making each query-seed cell a paired counterfactual across sources (demand draws and regime paths are action-independent given the seed, so exogenous randomness is exactly paired; histories diverge thereafter by design, and that divergence IS the treatment effect) — makes differences in U attributable to evidence selection. Varying only the consumer, with retrieval held constant, isolates consumption. An oracle ladder (Section 6) separates selection from veracity from interpretation from control.

The result. At $k = 3$, real retrieval reduces operating return relative to a no-retrieval control for LLM consumers (BM25 -103.5 $[-137, -71]$, dense -91.5 , hybrid -74.6 , rerank -43.6 ; all Holm-robust over the 105-arm family), while oracle-selected evidence from the same corpus adds up to $+178.9$ $[+135, +221]$. A no-text belief tuned only on development queries reaches 1881.1 ($+102.1$ over the prior, $p < 0.0002$): even the best constant policy that reads nothing beats every real-retriever arm, so the harm finding survives a tuned comparator. The loss decomposes cleanly: a relevance-to-factual refinement gap of 0.00 (both oracle sets saturate their consumer), an interpretation gap that shrinks $140.5 \rightarrow 36.4$ from naive to provenance-aware consumption, and a downstream control gap of $+456.8$ that dominates everything even under perfect semantics.

Contributions. (i) A frozen, regenerable retrieval→belief→control benchmark scoring evidence by sequential realised return, with retrieval quality on the same episodes (Section 4). (ii) A measured mismatch pattern — real retrieval harms, oracle evidence helps — with a tuned no-text comparator, a rule-vs-LLM interaction, and a four-gap attribution (Sections 5–6). (iii) A statistical discipline for this design: paired shared-seed worlds, crossed bootstrap, Holm over 105 arms, dual priors, text-cluster honesty check (Section 3.2). (iv) A reproducible artifact regenerating every table from stored episodes.

Scope. Synthetic feed and environments; constructed, not human-assessed, relevance (Section 7). No new retrieval model; no consumer or controller claimed preferable in general. The claim is about *evaluation* — score decision-bound evidence at induced actions and realised return, against a no-retrieval control and matched budget — scoped throughout to arms, priors and environments where corrections hold.

2 Related work

What we inherit. That relevance can diverge from downstream value is established: situational and task-based relevance critique [3]; eRAG’s weak relevance-to-utility correlation [30]; model-specific grounding utility [18]; distracting-passage harm that grows with retriever strength [7,33]; component-wise RAG evaluation with uncertainty [29,12]. That prediction quality differs from decision quality is established: task-based end-to-end learning [10,11,25], value-aware dynamics [13,21], decision calibration [34], and the classical value of information [17]. Probability quality is standardly assessed with proper scores and calibration diagnostics [15,8,26]. *None of these claims is ours:* each could have explained our results, and the controls below test those explanations rather than evading them. Closest are inventory decision-focused learning — where prior work trains *through* a stochastic program, we freeze an equivalent loop and vary only the supplied text — and the SPO loss [11], which already scores predictions by induced decision error and which our ΔJ applies to retrieved evidence; the closest operational-language system instead keeps its LLM around the solver as translator, not sensor [22].

What remains novel here. We are not aware of prior work combining a frozen sequential controller and world, retrieved natural-language evidence, matched budgets, paired world×language randomisation, and a six-node oracle ladder separating selection, veracity, interpretation and control. In particular, no prior study we know measures a rule-vs-LLM interaction on realised operating return, or a tuned no-text policy beating every tested real retriever while oracle evidence helps. The normalised value diagnostic is standard skill-score bookkeeping; we claim no new metric.

Environment and baselines. The controlled simulator is a single-item lost-sales inventory model; the secondary transfer uses a pinned published multi-echelon

package [1]. Retrieval baselines are Okapi BM25 [28] and cosine similarity over frozen sentence embeddings [27], scored with nDCG [19].

3 Framework

3.1 Objects and estimand

Let z be a latent operational regime, $\mathcal{D}$ a candidate pool of reports and q a fixed query expressing a standing information need. A retrieval system returns an ordered list; its top k documents are the evidence. A fixed consumer maps that evidence to a belief $b(z)$, and a fixed controller maps state and belief to an action.

Write $G_i(m)$ for the *realised* return of episode i under information source m , $G_i(m) = \sum_{t=0}^{T-1} r(s_t^{(i)}, a_t^{(i)}, z^{(i)})$, undiscounted over the fixed $T = 40$ horizon. Its expectation is estimated by the weighted benchmark return

$$J_w(m) = \sum_r w_r \operatorname{mean}_{f \in \mathcal{F}_r} \operatorname{mean}_{v \in \mathcal{V}_{r,f}} \operatorname{mean}_{i \in \mathcal{S}} G_i(m; r, f, v), \quad (2)$$

averaging seeds within a query v , queries within a family f , families within a regime r , and combining regimes with declared weights w_r . The *balanced* estimand uses uniform w_r and is primary; an *empirical* estimand matching the test bank mix (normal 0.2, delay 0.4, surge 0.4) is reported alongside every headline so that no conclusion rests on the weighting choice. Group sizes are unequal, so (2) is not an equal-row mean over episodes, and the difference is numerically material.

Effects are reported as paired differences ΔJ against a declared reference: for retrieval, a control that retrieves nothing; for consumers, the uninformed prior. Headline utility estimands use the 160 held-out test queries; development queries enter episodes only for matrix coverage. Gaps are signed differences of balanced J : retrieval $J(\text{actual}) - J(\text{NoInfo})$, relevance-to-factual $J(\text{OracleFactual}) - J(\text{OracleRelevant})$, interpretation $J(\text{PerfectBelief}) - J(\text{OracleFactual})$, control $J(\text{Hindsight}) - J(\text{PerfectBelief})$, with rungs defined in Section 6. A development-tuned constant no-text belief (Section 6.1) provides the stronger comparator. The normalised diagnostic SIVR = $(J(\text{condition}) - J(\text{NoInfo})) / (J(\text{oracle-semantic}) - J(\text{NoInfo}))$ is reported against both PerfectBelief and matched OracleFactual references with an explicit guard (NaN and a marked status, never absolute normalisation, when the reference gap is zero or negative); it is bookkeeping, not a new metric. With a fixed heuristic controller, the value of information need not be non-negative: our estimand is the value of information *to this controller class*, not universal value of information, so negative realised value diagnoses the pairing, never a violation of decision theory. The *perfect-semantic reference* — the true regime passed through the fixed controller — is demonstrably not an upper bound on realised return, so normalising by it does not yield a recovered fraction of anything; we report raw paired effects first and mark rather than interpret degenerate ratios.

3.2 Uncertainty

Queries are the primary unit; five paired seeds (60000–60004, disjoint from all development ranges) quantify environmental uncertainty, shared across every treatment. The design crosses worlds (seeds) with language (queries): every query is evaluated on the same seeds, so we resample one *shared* seed sample per regime plus queries with replacement (crossed bootstrap, $B = 5,000$; p -values are centred test inversions bounded below by $1/(B + 1)$, never zero). Paired randomisation testing follows IR significance-testing practice [32], and family-wise control follows classifier-comparison practice [9]. Semantic statistics use paired query bootstrap ($B = 10,000$). Multiplicity is controlled by Holm adjustment [16]: over the 105-arm utility family ($5 \text{ retrieval} \times 10 + 2 \text{ oracle} \times 10 + 7 \text{ interventions} \times 5$ — every (system, consumer, model, k) combination of the frozen matrix), and over ten preregistered headline semantic cells; all other contrasts are estimation with CIs, labeled exploratory. Three honesty provisions apply: (i) 200 IDs share 64 texts (63 test) — a text-cluster bootstrap (median $1.08\times$, max $1.19\times$ wider; cluster-bootstrap practice follows [4]) accompanies every starred claim, which holds under both; effective $N \approx 137$; (ii) inference is to new queries/seeds *within this bank*, not new language; (iii) regime-conditional statements are exploratory ($n = 5$ seeds).

4 Retrieval to semantic transfer

Setup: 200 queries over a 64-node report feed (40 dev / 160 test; regime-faithful pool judgments; test seal held until frozen analysis). Systems: random, BM25, instructed dense, RRF hybrid ($k = 120$), reranker over hybrid top-30 (depth dev-selected), oracle-relevant/factual ceilings, seven prespecified interventions. Consumers C0/C1/C3 (no C2 was ever defined): C0 (rule-based, model-free), C1 (naive LLM use), C3 (per-document extraction, frozen abstention $\tau = 0.5$, an arbitrary predeclared constant with no ablation; C1 prompts for regime probabilities plus event parameters as strict JSON, C3 for per-document entity, event, freshness, stance and confidence; a frozen-payload sweep over $\tau \in \{0.3, 0.7\}$ leaves C3 accuracy and abstention unchanged to four decimals, since per-document support avoids the threshold band (identical beliefs imply identical downstream returns, so no utility rerun is needed); abstentions carry the prior, score 0 accuracy, keep prior Brier — a reject option in the classical sense [5], without a coverage-risk analysis [14], which we leave to future work). Two pinned open-weight models (8B/14B); 11,018 content-addressed cache files, zero assembly writes.

Table 1 shows the frozen ranker is point-ordered (BM25 0.115 to reranked 0.183; dense/hybrid/rerank intervals overlap), though rerank’s edge over hybrid shrinks out of sample ($+0.024 \text{ dev} \rightarrow +0.001 \text{ test}$) and trades recall for nDCG — a finding, not a tuning trigger. No position analysis beyond $k \in \{3, 5\}$ is attempted; $k = 5$ repeats the pattern (all ten C1 arms Holm-robust negative) with full numbers in the frozen artifact.

Semantic accuracy (Table 2) sits at 0.275–0.450 outside abstention-collapse cells — barely above model-free C0 on identical evidence and far below oracle

Table 1. Retrieval quality on the 160 test queries (trec_eval; 95% query-bootstrap CIs: BM25 [0.095, 0.135], dense [0.153, 0.198], hybrid [0.157, 0.207], rerank [0.156, 0.210] — point-ordered with overlapping intervals). The frozen ranker is RRF over BM25 and instructed dense embeddings ($k = 120$) with reranker depth 30, selected on dev only.

System	nDCG@10	R@10	R@20	RR
BM25	0.1150	0.1365	0.2604	0.2102
Dense	0.1754	0.2062	0.3771	0.2656
Hybrid RRF	0.1817	0.2167	0.4094	0.2857
Reranked	0.1829	0.1938	0.3708	0.2812

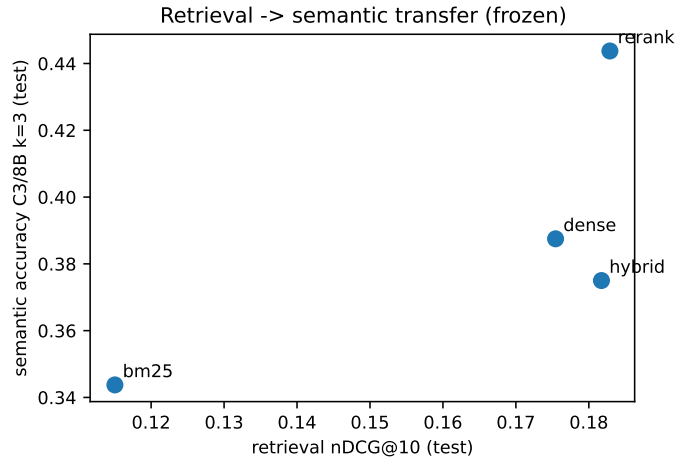


Fig. 1. Retrieval quality against semantic quality across the four practical rankers (test queries, C3/8B, $k = 3$). The ordering transfers in rank while absolute accuracy stays low — the pattern the utility stage then breaks.

evidence (C3: 0.96–0.97). C1 is *worse* than C0 in 17 of 20 real-retriever cells (BM25/8B: 0.275 vs 0.406). C3 beats C1 on oracle (+0.25) but marginally on real retrieval, and abstains on garbage. Scale shows no systematic advantage (34/42 nulls, null = CI includes zero; 8 scattered mixed-sign cells, max $|\Delta| = 0.144$). Better retrieval is directionally consistent with better semantics for LLM consumers but underpowered ($n = 4$, exact $p = 0.083$ –0.92); orderings are stable across consumers ($\rho \geq 0.80$, $p < 0.003$). Evidence-hit explains +0.36–+0.42 accuracy on retrieval systems. Held-out entities score *higher* (BM25/C1/8B 0.38 vs 0.23; descriptive, plausibly an easier mix).

The retriever $\times$ consumer interaction (Table 3) is carried by rule-vs-LLM contrasts (3/10 Holm-robust); C1-vs-C3 and parity differences retain nulls — a level shift between model-free and LLM consumption, not between LLM consumers.

Table 2. Semantic quality on test queries at $k = 3$ (160 queries). C0 is the deterministic rule-based reference (model-free); C1 naive LLM RAG; C3 provenance-aware with abstention. Abstention rates: C3/random 0.56–0.94, else ≤ 0.03 . Per-cell 95% query-bootstrap CIs (median half-width 0.07) are tabulated in the frozen artifact.

(a) Regime accuracy					
System	C0	C1/8B	C1/14B	C3/8B	C3/14B
BM25	0.406	0.275	0.338	0.344	0.338
Dense	0.412	0.287	0.281	0.388	0.331
Hybrid	0.425	0.362	0.350	0.375	0.350
Rerank	0.400	0.406	0.412	0.444	0.444
Random	0.350	0.237	0.225	0.081	0.006
Oracle	0.594	0.700	0.719	0.963	0.969

(b) Brier score					
System	C0	C1/8B	C1/14B	C3/8B	C3/14B
BM25	0.254	0.356	0.314	0.271	0.288
Dense	0.233	0.329	0.319	0.260	0.282
Hybrid	0.260	0.315	0.291	0.263	0.273
Rerank	0.297	0.258	0.257	0.269	0.262
Random	0.286	0.336	0.327	0.252	0.226
Oracle	0.157	0.121	0.127	0.028	0.035

5 Sequential utility

Feeding the exact frozen beliefs through the frozen controller and simulator (105 arms $\times$ 200 queries $\times$ 5 paired seeds = 105,000 episodes for coverage; headline estimands use the 160 test queries) gives Table 4 and Figure 2.

All BM25/dense/hybrid/rerank LLM-consumer arms sit at or below the no-retrieval control at $k = 3$, with Holm-robust harm for all C1 arms (−41 to −104) and three of eight C3 cells (14B only); every real-retriever C0 comparison is directionally negative but Holm-n.s. A majority of individual queries are harmed (46–74% across real-ranker arms and budgets, seed-averaged $\Delta < 0$, vs 3.7% for oracle-C3). A second fixed policy class (belief-adaptive base-stock, same beliefs/seeds/protocol, labeled extension) preserves rerank/C1 (−55.0*) and rerank/C3 (−94.8*) harm and oracle-LLM gains (+100–+107*), but reverses BM25/C1 (+54.5*) and the tuned constant (−39.3): harm lives at the pairing level, which is what decision-bound evaluation must score. Under empirical weights, BM25/dense/hybrid C1 harm and all oracle-LLM gains persist Holm-robust; rerank-C1/C3 and all real-retriever C0 comparisons go Holm-n.s. (neutral arms flip sign), so universality is disclaimed and only the robust subset is headlined. Within naive consumption, harm generally declines with test nDCG (8B: −104 $\rightarrow$ −92 $\rightarrow$ −75 $\rightarrow$ −44; 14B: −90 $\rightarrow$ −93 $\rightarrow$ −74 $\rightarrow$ −41 with one adjacent swap) — against a pure weak-retriever artifact, which would not grade smoothly with ranking quality while the oracle flips sign. Better IR is less

Table 3. Retriever $\times$ consumer interaction (rerank vs BM25, $k = 3$, accuracy): $\Delta\Delta$ with 95% CI and Holm p over 10 preregistered cells. Q5/Q6 rows report Holm p only (Q6-C3 degenerate: identical profiles).

Cell	$\Delta\Delta$ [95% CI]	Holm p
Q3_rerank-bm25_C0-C1_k3_8B	−0.138 [−0.225, −0.056]	0.0150*
Q3_rerank-bm25_C0-C3_k3_8B	−0.106 [−0.175, −0.037]	0.0360*
Q3_rerank-bm25_C1-C3_k3_8B	+0.031 [−0.050, +0.113]	1.0000
Q3_rerank-bm25_C0-C1_k3_14B	−0.081 [−0.150, −0.013]	0.1855
Q3_rerank-bm25_C0-C3_k3_14B	−0.113 [−0.188, −0.037]	0.0297*
Q3_rerank-bm25_C1-C3_k3_14B	−0.031 [−0.087, +0.025]	1.0000
Q5_rerank_C3-C1_k3_8B	+0.037 [−0.019, +0.094]	1.0000
Q5_rerank_C3-C1_k3_14B	+0.031 [−0.019, +0.081]	1.0000
Q6_rerank_8B-14B_C1_k3	−0.0063 [−0.0312, +0.0187]	1.0000
Q6_rerank_8B-14B_C3_k3	+0.0000 [+0.0000, +0.0000]	1.0000

harmful for naive consumption (rerank > hybrid > dense $\approx$ BM25 for C1) but still harmful or neutral — except random/C3/14B +11.0: C3 abstains on random evidence and falls back to the strong prior, so this cell measures the fallback, not retrieval helping; kept out of headlines. (C3 already implements the selective gate — abstention maps to the prior by construction, verified by a no-op selective rerun — so no separate selective arm exists to run.)

6 Locating the loss

The ladder (Table 5, Figure 3) attributes the loss rung by rung with controller, world, seeds and budget fixed: NOINFO (prior), ACTUALRETRIEVAL (frozen belief row), ORACLERELEVANT (top- k by grade), ORACLEFACTUAL (+fresh, entity-matching, non-refuting), PERFECTBELIEF (true-regime one-hot, retrieval-independent), HINDSIGHTORACLE (clairvoyant MILP) — separating selection, veracity, interpretation and control: the relevance-to-factual refinement gap is 0.00 in every arm (both oracle sets saturate their consumer identically); the interpretation gap shrinks 140.5 $\rightarrow$ 36.4 from naive to provenance-aware consumption; the downstream control gap (+456.8, bundling demand-foresight value with controller suboptimality) dominates everything, so even perfect semantics leave more than twice the perfect-semantics value on the table. Regime slices (exploratory, per-regime paired CIs, no multiplicity control) show the mechanism is concentrated, not diffuse: reranked evidence helps substantially in delay worlds (+150–+206 across consumers and models) while hurting in normal (−225–−157) and surge (−129–−79) worlds.

Interventions on oracle evidence. Perturbing the oracle-factual base seven prespecified ways (drop-decisive, stale-swap, inject-contradiction, wrong-entity-swap, duplicate-top, order-reverse, irrelevant-inject) stays positive throughout (duplicate-top/C3 +185 balanced / +215 empirical; all Holm-robust): excellent evidence stays

Table 4. Paired ΔJ vs no-retrieval control at $k = 3$ (160 test queries $\times$ 5 seeds; crossed bootstrap, $B = 5,000$). *: Holm rejection (105 arms); unstarred cells Holm-n.s. Displayed CIs are unadjusted; stars carry multiplicity control. A harm claim is headlined only if Holm-robust under balanced weights and same-signed under empirical weights. Cluster CIs (median $1.08\times$, max $1.19\times$) agree in direction for every starred cell. Refs: NoInfo 1779.0, PerfectBelief 1994.4, Hindsight 2451.2, NoText_Tuned 1881.1 (§6.1).

System	C0	C1/8B	C1/14B	C3/8B	C3/14B
	−8.5	−103.5*	−89.7*	−43.4	−59.8*
BM25	[−32, +16]	[−137, −71]	[−118, −63]	[−71, −15]	[−82, −37]
	−4.6	−91.5*	−93.2*	−27.9	−55.2*
Dense	[−26, +17]	[−122, −61]	[−121, −66]	[−58, +3]	[−79, −31]
	−9.8	−74.6*	−73.9*	−36.6	−54.4*
Hybrid	[−35, +15]	[−105, −44]	[−97, −51]	[−64, −10]	[−73, −35]
	−40.2	−43.6*	−40.9*	−21.8	−20.7
Rerank	[−68, −13]	[−65, −23]	[−62, −20]	[−56, +11]	[−48, +7]
	−35.0	−90.4*	−89.9*	−13.1	+10.9*
Random	[−62, −9]	[−110, −71]	[−104, −75]	[−26, −2]	[+7, +14]
	+39.3	+74.9*	+70.6*	+178.9*	+162.5*
Oracle	[+9, +69]	[+38, +111]	[+32, +108]	[+135, +221]	[+125, +200]

excellent. Contra-injection splits the consumers (C1 +29n.s. vs C3 +165*) — provenance-awareness pays, but only where evidence starts near-perfect.

6.1 The tuned no-text comparator

Because a fixed prior is a weak comparator, we tuned a constant no-text belief on development queries only (66-triple grid at step 0.1, 40 queries, disjoint seeds; selection, not inference) and evaluated it once on test: always anticipate surge, with runners-up (0.1, 0, 0.9) and (0, 0.1, 0.9) at dev $J = 1869.5$ behind the winner’s 1885.6 (full grid in the frozen artifact). The top-3 dev order is preserved on test (runners-up $J = 1869.1$ vs winner 1881.1; paired winner-minus-runner-up +19.8 [+4, +20], exploratory outside Holm), against winner’s-curse luck. It reaches $J = 1881.1$, +102.1 over the prior ($p < 0.0002$ crossed bootstrap, $B = 5,000$, same paired scheme; Holm over the 9 tuned-family comparisons — tuned-vs-prior plus tuned-vs-each-real-C1/C3-arm — rejects 9/9; labeled extension, SIVR 0.474, i.e. 47% of the perfect-semantics value recovered) — and still every real-retriever arm sits below it (paired tuned-minus-arm CIs, all excluding zero: rerank/C1 [+108, +184], rerank/C3 [+82, +165], oracle-C3 [−118, −35] the other way). The comparator cuts both ways: it rules out a straw-man prior, but its edge is prior-dependent (lost at −39 normal-heavy) and mechanistic (equals perfect semantics in surge worlds since a missed unit costs 18 against ~ 1 –3 to carry one).

A secondary transfer to a published multi-echelon simulator corroborates no-text robustness only: no arm beats NoInfo there (perfect beliefs −50.7 [−62, −41]) while rankings flip (rerank −240.7), reported as environment dependence given the transfer’s own two-class projection artifact.

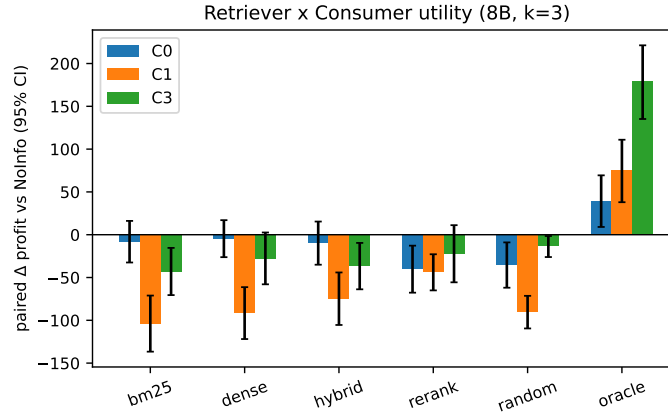


Fig. 2. Paired change in operating return vs retrieving nothing (8B, $k = 3$, 95% crossed intervals). Real retrieval sits at or below zero for LLM consumers; only oracle evidence crosses into positive value.

7 Boundaries and limitations

The feed is synthetic, single-SKU and finite-regime, with constructed, not human-assessed, relevance — unlike the situational programmes that motivate this work, which require real users (validation unexecuted; agent-audit priors 0.82/0.90, $\kappa = 0.874$, are priors, not substitutes); absolute nDCG values should not be compared outside this bank. One simulator; two fixed controller classes (primary LP plus a base-stock extension that reverses some arms), two open-weight models from one family; the C0 pattern needs no LLM and carries the model-free half of the claim. 160 test queries over 63 unique texts (effective $N \approx 137$); five seeds adequate for balanced headlines (query:seed variance 6–7:1 in pilots; leave-one-seed-out headline Δ within ± 10), not for regime-conditional inference (exploratory). Held-out templates unavailable; held-out entities score higher for an untested mix reason. The tuned comparator is prior-dependent (loses under normal-heavy weights, where BM25/C1 is $-30 [-93, +35]$ n.s. and oracle-C3 $+79 [+6, +151]$); BM25/dense/hybrid C1 harm and oracle-LLM value point the same way under all five weightings checked (full CIs under balanced and empirical only). SIVR is reported bookkeeping with signed guards, not a new metric. Nothing here establishes deployment safety.

8 Conclusion

For a decision consumer, evidence quality must be scored at the level of induced actions and realised return: better-ranked evidence yields marginally better beliefs yet, for LLM consumers, operating returns at or below reading nothing on real retrieval (Holm-robust harm on every real-retriever naive-use arm), while

Table 5. Four-gap decomposition at $k = 3$ (rerank arm; balanced J ; paired crossed 95% CIs inline). Rel→fact is 0.00 in every arm (both oracle sets saturate identically). Control +456.8 [+387, +538] is retrieval-independent (Hindsight–PerfectBelief) and shown once.

Arm	Retrieval	Rel→Fact	Interp.
	−40.2	+0.0	+176.0
C0	[−68, −13]	[+0, +0]	[+130, +220]
	−43.6	+0.0	+140.5
C1/8B	[−65, −23]	[+0, +0]	[+108, +172]
	−21.8	+0.0	+36.4
C3/8B	[−56, +11]	[+0, +0]	[+17, +58]
	−40.9	+0.0	+144.8
C1/14B	[−62, −20]	[+0, +0]	[+114, +177]
	−20.7	+0.0	+52.9
C3/14B	[−48, +7]	[+0, +0]	[+31, +77]
Control (all arms)	+456.8 [+387, +538]		

oracle evidence from the same corpus is worth up to +179. The failure is a mismatch between retrieval, consumer and controller — a second fixed policy class preserves rerank harm and oracle gains while reversing BM25/C1 and the tuned constant, confirming harm lives at the pairing level — located by the ladder: selection is not the bottleneck (relevance-to-factual 0.00), interpretation closes part of the gap (141 → 36), and control leaves the most value unrealised (+456.8; this bundles demand-foresight value with controller suboptimality, since the clairvoyant resolves sample-path randomness the regime controller never sees). Score retrieval intended for a decision loop against that loop, with a no-retrieval control, a tuned no-text comparator and a matched budget — and do not assume a relevance gain survives the handoff.

A Design, accounting and reproduction

Environment, costs and timing. Single-item lost-sales inventory over $T = 40$ periods: revenue \$10/sale, ordering \$5 fixed plus \$2/unit, holding \$1, stockout penalty \$8, initial on-hand 30, Poisson demand (mean 8). Regimes: NORMAL; SUPPLIER DELAY (lead time up in window); DEMAND SURGE (demand rate up). Within a period: receive arrivals, realise demand, fulfil, order. Orders placed at t under lead time L arrive at $t + L$; in-transit orders keep arrival periods. One shared schedule implementation.

Information release and belief to action. Warning at period 12; events periods 20–29. Before publication all controllers plan under the benchmark prior; after, the interpreted belief replaces it with state and pipeline carried over. The controller solves a receding-horizon LP per period (SciPy `linprog`, HiGHS; flow-balance and bound constraints with horizon 40 in the artifact), executing its first order by minimising $\sum_{\tau} (2\hat{q}_{\tau} + \hat{I}_{\tau} - 18\hat{s}_{\tau})$, where the coefficients are

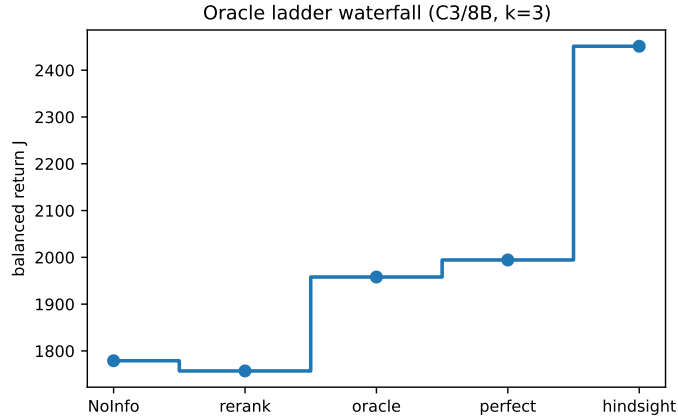


Fig. 3. Oracle ladder waterfall (C3/8B, $k = 3$, balanced J): actual retrieval destroys value vs no retrieval, oracle evidence restores it, and the hindsight reference shows control leaving the largest gains unrealised even under perfect semantics.

the stated variable/holding costs and the revenue-plus-stockout at stake, under belief-weighted demand and lead time, identically across sources. The online program is a continuous relaxation: the simulator’s \$5 per-order fixed cost is charged on execution but not anticipated in planning (uniform across arms); only the hindsight reference solves the mixed-integer form with setup binaries, so the control gap further bundles relaxation against foresight.

Retrieval, consumers, models. BM25 ($k_1 = 1.5$, $b = 0.75$, idf floored at zero); instructed dense embeddings; RRF hybrid ($k = 120$) [6]; reranker over hybrid top-30 (depth 30 selected on dev only); oracle-relevant and oracle-factual ceilings; seven prespecified evidence interventions; budgets $k \in \{3, 5\}$. C0 rule-based reference; C1 naive LLM use; C3 per-document extraction with frozen abstention ($\tau = 0.5$). LLM consumers use `Qwen/Qwen3-8B-AWQ@4da05a8e` and `Qwen/Qwen3-14B-AWQ@31c69efc`, resolvable revisions at `huggingface.co`, served once via frozen prompts (SHA C1 66e6890b, C3 5966cadd); 11,018 outputs cached content-addressed with prompt/model SHAs and usage records; kernels ran qrel-free with hard-abort seal guards.

Accounting and reproduction. 25,200 frozen belief rows (2 models $\times$ C0/C1/C3) with SHA manifest (0 gate errors); 108,000 controlled episodes plus 107,000 gym episodes, seeds 60000–60004 throughout; 3,424-file superseded GPT partial preserved but excluded under a logged model-substitution amendment (billing block; replacement probes gated in the artifact log). Cross-session cache nondeterminism (36 disputed keys, 140 rows) moves headline effects by at most 0.0021 Brier / 0.0 accuracy. Every table regenerates from stored episodes and beliefs (57/57 byte-identical rerun). A synthetic study does not establish deployment safety.

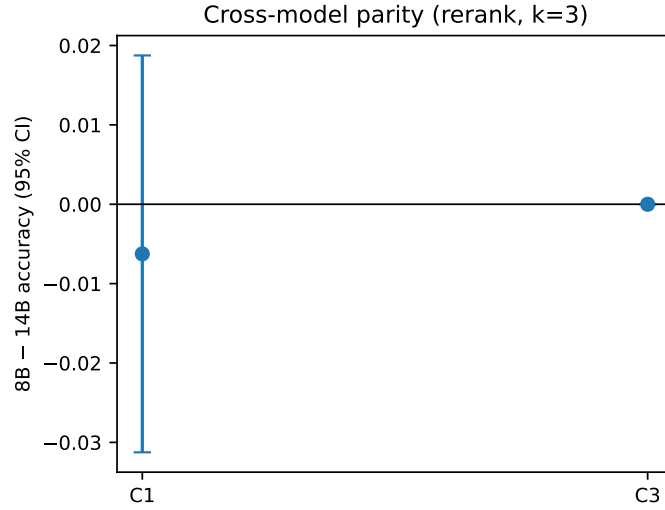


Fig. 4. Cross-model accuracy parity (rerank, $k = 3$; 8B minus 14B with 95% intervals). Within-family scale does not move semantic quality; the transfer pattern replicates across both models.

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